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Evaluation of the applicability of state estimation filters for indoor robot positioning

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Abstract– **Indoor robot positioning plays an important role in ensuring autonomous navigation in smart environments where Global Positioning System (GPS) signals are unavailable. Ultra-Wideband (UWB) technology offers high accuracy and multipath interference resistance, providing a suitable foundation for indoor positioning systems. Therefore, this study focuses on evaluating the performance of three state estimation algorithms—the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and the Particle Filter (PF)—in handling nonlinear models of positioning systems based on UWB. The results show that the EKF has advantages in processing speed and low computational cost, making it suitable for real-time systems, though it remains sensitive to noise and highly nonlinear models. The UKF demonstrates a balance between estimation accuracy and computational efficiency due to its ability to provide reliable estimates under nonlinear conditions without increasing computational cost. The PF offers high stability and the ability to handle non-Gaussian noise; however, it requires significant computational resources, especially when the number of particles is large.**

Keywords– **UWB, positioning, localization, tracking, Kalman filter, indoor robot positioning.**

1 Introduction

In the context of rapidly growing connectivity technologies, the demand for accurate and reliable Indoor Positioning Systems (IPS) has become increasingly prominent. Global Navigation Satellite Systems (GNSS) [1] perform well outdoors, but their signals degrade severely in enclosed spaces, so they cannot meet indoor positioning requirements. UWB [2, 9] has emerged as a promising solution due to centimeter-level accuracy and strong resistance to interference and multipath, outperforming alternatives such as Wi-Fi, Bluetooth Low Energy (BLE), or Radio Frequency Identification (RFID) [3].

To fully exploit UWB for robot positioning, robust state-estimation algorithms are required. The EKF, UKF, and PF [4] are three widely used approaches that can address nonlinearities that standard Kalman Filter cannot handle.

This paper compares the EKF, UKF, and PF for UWB-based indoor robot positioning using criteria such as accuracy, computational cost, and stability. The results support the selection of appropriate algorithms for low-cost and reliable indoor positioning systems.

This work has three main contributions. Firstly, we build an indoor positioning system consisting of four fixed reference nodes (Anchors) and one mobile target (Tag) using UWB modules with an integrated two-way ranging (TWR) distance measurement method. Secondly, we talk about implement an anchor-wise error calibration method to compensate for systematic ranging errors of each UWB anchor, thereby improving the reliability of the measured distances. Thirdly, we implement the EKF, UKF, and PF to process the range measurements from the four anchors to the robot-mounted tag and evaluate the positioning accuracy under different initialization settings.

The remainder of this paper is organized as follows. Section I presents the context, motivation, and objectives of the study. Section II describes the model and architecture of the positioning system. Section III presents the principles and implementation of the EKF, UKF, and PF in the proposed system. Section IV focuses on experimental design and performance evaluation. Finally, Section V concludes the article and discusses future directions.

2 Indoor positioning system model

2.1 System overview

2.1.1 UWB system: The positioning system is built using compact DWM1001 modules integrating a UWB transceiver, an ARM Cortex-M4 processor, and BLE. The modules are configured as fixed reference nodes (Anchors) and a mobile target (Tag): four anchors are deployed at known locations and a mobile robot is equipped with a UWB tag. The complete system model is shown in Fig. 1. The anchors and tag perform continuous TWR [5] by exchanging timestamped packets to estimate distances from time-of-flight information. All anchors interact with the tag simultaneously, enabling robust range measurements even in dynamic environments. In each ranging cycle, the tag receives distance measurements from four anchors, which are then used as inputs for position estimation.

Both the tag and the anchors run the *DWM1001_PANS_R2.0.hex* firmware, which supports a ranging update rate of up to 160 Hz, enabling fast distance measurements suitable for real-time mobile robot positioning. With this high update frequency and appropriate estimation methods, the system can achieve high positioning accuracy.

2.1.2 Main processing unit: The hardware includes a four-wheel indoor mobile robot equipped with a Raspberry Pi 4 as the main processing unit. The Pi reads continuous range data from the tag via a serial connection. The EKF, UKF, and PF are implemented as separate Python modules and executed directly on the Pi 4.

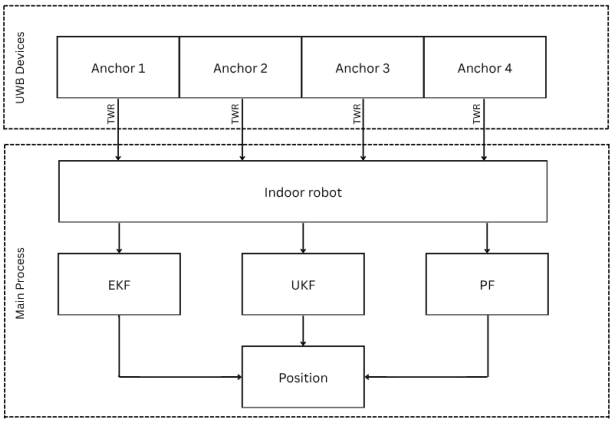

Figure 1. System architecture.

2.2 Calibration model

UWB range can achieve centimeter-level accuracy and a maximum measurable range of more than 100 m in outdoor environments. However, both of them have been significantly degraded when UWB is operated in indoor environments. Therefore, the preliminary calibration is necessary to use UWB for indoor environments. The UWB calibration procedure for distance measurements using the Symmetric Double-Sided Two-Way Ranging (SDS-TWR) method is bias calibration.

We assumed that the relationship between measured distance, what we called d, and the actual distance, d', can be espress as equation follow:

$$d' = md + c \tag{1}$$

where m and c denotes the paraments of the bias calibration which can be obtained by taking multiple measurements at different distances [6].

2.3 Mathematical model

When initializing the system, we assume that the initial velocity components are zero, i.e., $v_{x,0} = v_{y,0} = 0$.

The system uses four fixed anchors and one mobile tag. The coordinates of the i-th anchor are denoted by $\mathbf{a}_i = (x_{a_i}, y_{a_i})^\top$, for $i = 1, \ldots, 4$. The measurements are distances from the tag to each anchor: $z_{i,k}$ is the measurement from the tag to anchor i at sample k, and the measurement vector at time k is $\mathbf{z}_k = [z_{1,k}, \ldots, z_{4,k}]^\top$.

The sampling interval between two consecutive samples is determined from the tag update characteristics; in this system, $dt = 0.1\,\mathrm{s}$. The process-noise matrix $\mathbf{Q}$ and measurement-noise matrix $\mathbf{R}$ are assumed under different cases to analyze convergence. The values of σ_a^2 are adjusted in different experimental scenarios to study filter response to tag dynamics and sensitivity to noise levels.

2.3.1 State model: We choose a 2D constant-velocity (CV) model with state:

$$\mathbf{x_k} = \begin{bmatrix} x_k \\ y_k \\ v_{x,k} \\ v_{y,k} \end{bmatrix}. \tag{2}$$

The discrete-time dynamics are linear:

$$\mathbf{x}_{k+1} = \mathbf{F}\,\mathbf{x}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}). \tag{3}$$

The state-transition matrix $\mathbf{F}$ (with sampling period dt) is

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & dt & 0 \\ 0 & 1 & 0 & dt \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

If the process noise is modeled as zero-mean white acceleration with variance σ_a^2, the discrete-time process noise covariance matrix for the constant-velocity (CV) motion model is given by

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{dt^3}{3} & 0 & \frac{dt^2}{2} & 0 \\ 0 & \frac{dt^3}{3} & 0 & \frac{dt^2}{2} \\ \frac{dt^2}{2} & 0 & dt & 0 \\ 0 & \frac{dt^2}{2} & 0 & dt \end{bmatrix}, \tag{5}$$

where dt denotes the sampling interval and σ_a^2 represents the variance of the continuous-time acceleration noise.

The CV model assumes that acceleration is not explicitly included in the state vector but is treated as a random disturbance. The structure of $\mathbf{Q}$ in (5) results from integrating white acceleration noise over the sampling interval dt, which propagates uncertainty into both position and velocity states. In practical tuning, only the scalar parameter σ_a^2 is adjusted, while the matrix structure remains unchanged.

2.3.2 Measurement model: Since DWM1001 provides direct distances (TOA/TWR) from the tag to each anchor, the nonlinear measurement model is

$$z_{i,k} = h_i(\mathbf{x}_k) + v_{i,k}, \qquad v_{i,k} \sim \mathcal{N}(0, \sigma_{r,i}^2), \tag{6}$$

with measurement function

$$h_i(\mathbf{x}_k) = \sqrt{(x_k - x_{a_i})^2 + (y_k - y_{a_i})^2}. \tag{7}$$

The measurement vector is

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{h}(\mathbf{x}) = \begin{bmatrix} h_1(\mathbf{x}) \\ \vdots \\ h_m(\mathbf{x}) \end{bmatrix}, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{8}$$

where $\mathbf{R}$ is diagonal and $\sigma_{r,i}^2$ is the assumed variance of the range measurement error to anchor i:

$$\mathbf{R} = \mathrm{diag}(\sigma_{r,1}^2,\ \sigma_{r,2}^2,\ \ldots,\ \sigma_{r,m}^2). \tag{9}$$

3 Positioning algorithm model

3.1 Least-squares estimation (LSE)

LSE is used to estimate the initial coordinates from the first four TWR measurements in order to initialize the EKF, UKF, and PF with a reliable starting state, improving convergence speed. The LSE formulation [7] is

$$\mathbf{A}\mathbf{x}_{\mathrm{LS}} = \mathbf{b} \tag{10}$$

where

$$\mathbf{A} = \begin{bmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \\ \vdots & \vdots \\ x_n - x_1 & y_n - y_1 \end{bmatrix}, \quad \mathbf{x}_{\mathrm{LS}} = \begin{bmatrix} x_{\mathrm{LS}} \\ y_{\mathrm{LS}} \end{bmatrix}$$

$$\mathbf{b} = \frac{1}{2}\begin{bmatrix} x_2^2 + y_2^2 - d_2^2 - (x_1^2 + y_1^2 - d_1^2) \\ x_3^2 + y_3^2 - d_3^2 - (x_1^2 + y_1^2 - d_1^2) \\ \vdots \\ x_N^2 + y_N^2 - d_N^2 - (x_1^2 + y_1^2 - d_1^2) \end{bmatrix} \tag{11}$$

The LSE estimate is computed by solving the linear system.

$$\mathbf{x}_{\mathrm{LS}} = (\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b}. \tag{12}$$

3.2 Filter implementations

3.2.1 Extended Kalman Filter: EKF extends the Kalman filter to nonlinear systems by first-order linearization using a Taylor expansion. For UWB positioning, the nonlinearity mainly appears in the measurement model. The state model and measurement model are initialized as above, and the initial state is set from the LSE estimate.

The EKF linearizes the measurement function $h(\mathbf{x}_k)$ by computing the Jacobian

$$\mathbf{H_k} = \frac{\partial h}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_{k|k-1}}. \tag{13}$$

For UWB positioning, the Jacobian of the measurement function is

$$\mathbf{H_k}[i,:] = \begin{bmatrix} \frac{\hat{x}_{k|k-1} - x_{a_i}}{\hat{d}_{i,k}} & \frac{\hat{y}_{k|k-1} - y_{a_i}}{\hat{d}_{i,k}} & 0 & 0 \end{bmatrix}, \tag{14}$$

where $\hat{d}_{i,k} = \sqrt{(\hat{x}_{k|k-1} - x_{a_i})^2 + (\hat{y}_{k|k-1} - y_{a_i})^2}$ is the estimated distance from the tag to anchor i.

3.2.2 Unscented Kalman Filter: UKF addresses nonlinearities via the unscented transform, avoiding explicit derivative computation as in the EKF. The UKF uses a carefully chosen set of *sigma points* to represent the state distribution. With state dimension $n_x = 4$, the UKF uses

Algorithm 1: EKF

Input : Range measurements $d_1, \ldots, d_4$; anchor positions; sampling interval Δt

Initialization: State: $\mathbf{x} = [x, y, v_x, v_y]^\top$ (initial position from LSE, initial velocity assumed zero)
Transition matrix: $\mathbf{F}(\Delta t)$
Initial covariance: $\mathbf{P}_0$ (tunable)
Process noise : $\mathbf{Q}$
Measurement noise : $\mathbf{R}$

while *measurement* $\mathbf{z}$ *available* **do**
  if *not init* **then**
    initialize $\mathbf{x}$ by LSE; set $v = 0$
  `// predict`
  $\mathbf{x} \leftarrow \mathbf{F}\mathbf{x}$; $\mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}$
  `// update`
  Evaluate $\mathbf{H}$ at $\mathbf{x}$; $\hat{\mathbf{z}} = \mathbf{h}(\mathbf{x})$ $\mathbf{S} \leftarrow \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}$;
  $\mathbf{K} \leftarrow \mathbf{P}\mathbf{H}^\top \mathbf{S}^{-1}$ $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}})$;
  $\mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}$

Output: $[x, y]$

$2n_x + 1 = 9$ sigma points:

$$\begin{aligned} \chi_k^{(0)} &= \hat{\mathbf{x}}_{k|k} \\ \chi_k^{(i)} &= \hat{\mathbf{x}}_{k|k} + \left(\sqrt{(n_x + \lambda)P_{k|k}}\right)_i, \quad i = 1, \ldots, n_x \\ \chi_k^{(i)} &= \hat{\mathbf{x}}_{k|k} - \left(\sqrt{(n_x + \lambda)P_{k|k}}\right)_{i-n_x}, \quad i = n_x + 1, \ldots, 2n_x \end{aligned} \tag{15}$$

The parameter $\lambda = \alpha^2(n_x + \kappa) - n_x$, where α controls the sigma-point spread (we use 10^{-1}) and $\kappa = 1.0$ is kept positive to ensure positive definiteness and avoid overly wide sigma-point dispersion.

The sigma-point weights are

$$\begin{aligned} w_m^{(0)} &= \frac{\lambda}{n_x + \lambda} \\ c^{(0)} &= w_m^{(0)} + (1 - \alpha^2 + \beta) \\ w_m^{(i)} = w_c^{(i)} &= \frac{1}{2(n_x + \lambda)}, \quad i = 1, \ldots, 2n_x \end{aligned} \tag{16}$$

with β incorporating prior information (set to $\beta = 2$ for Gaussian distributions).

UKF is particularly effective for UWB positioning systems due to its ability to handle nonlinearities without requiring derivative computations while maintaining high accuracy even under strong nonlinear dynamics.

3.2.3 Particle Filter: PF is a Monte Carlo filtering approach using a set of particles to approximate the posterior distribution. Each particle represents a hypothesis about the system state and is assigned a weight reflecting the likelihood of that hypothesis.

The posterior distribution is approximated as

$$p(\mathbf{x}_k | \mathbf{z}_{1:k}) \approx \sum_{i=1}^{N} w_k^{(i)} \delta(\mathbf{x}_k - \mathbf{x}_k^{(i)}), \tag{17}$$

where N is the number of particles, $\mathbf{x}_k^{(i)}$ is the state of

Algorithm 2: UKF

Input : Range measurements $d_1, \ldots, d_4$; anchor positions; sampling interval Δt

Initialization: State: $\mathbf{x} = [x, y, v_x, v_y]^\top$ (initial position from LSE, initial velocity assumed zero)
Initial covariance: $\mathbf{P}_0$ (tunable)
Sigma-point parameters: α, β, κ
Process noise : $\mathbf{Q}$
Measurement noise : $\mathbf{R}$

while *measurement* $\mathbf{z}$ *available* **do**
  if *not init* **then**
    initialize $\mathbf{x}$ by LSE; set $v = 0$
  `// predict`
  Form sigma points; propagate via $\mathbf{f}(\cdot, \Delta t)$; recover $\mathbf{x}^-, \mathbf{P}^-$; $\mathbf{P}^- \leftarrow \mathbf{P}^- + \mathbf{Q}$
  `// update`
  Propagate through $\mathbf{h}(\cdot)$; compute $\hat{\mathbf{z}}, \mathbf{P}_{zz} + \mathbf{R}, \mathbf{P}_{xz}$ $\mathbf{K} \leftarrow \mathbf{P}_{xz}(\mathbf{P}_{zz} + \mathbf{R})^{-1}$
  $\mathbf{x} \leftarrow \mathbf{x} + \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}})$; $\mathbf{P} \leftarrow \mathbf{P} - \mathbf{K}\mathbf{P}_{zz}\mathbf{K}^\top$

Output: $[x, y]$

particle i, and $w_k^{(i)}$ is its weight.

Weights are updated using the measurement likelihood:

$$w_k^{(i)} \propto w_{k-1}^{(i)} p(\mathbf{z}_k | \mathbf{x}_k^{(i)}),$$

assuming Gaussian range errors:

$$p(\mathbf{z}_k | \mathbf{x}_k^{(i)}) = \prod_{j=1}^{4} \mathcal{N}(z_{j,k}; \hat{z}_{j,k}^{(i)}, \sigma_{r,j}^2),$$

with $\hat{z}_{j,k}^{(i)} = \sqrt{(x_k^{(i)} - x_{a_j})^2 + (y_k^{(i)} - y_{a_j})^2}$.

PF provides a flexible solution for UWB positioning, especially in complex multipath environments. However, PF requires significant computational resources, and the number of particles must be tuned to balance accuracy and real-time performance.

4 Experimental design and evaluation

4.1 Bias Calibration

The experiments were carried out at fixed distances from 1.2 m to 5.4 m in LOS environment, with 0.6 m increments. For each distance, 5000 measurement samples were collected to compute the RMSE, and calibration was performed using the method described in Section 2.

To evaluate the effectiveness of the proposed calibration method, the measured data are recalibrated using the estimated parameters m and c for each anchor, and the results are visualized through RMSE graph. Moreover, the cumulative distribution function (CDF) of LSE before and after calibration is also presented to evaluate the impact of the proposed method on the overall system performance.

Algorithm 3: Particle Filter (PF)

Input : Range measurements $d_1, \ldots, d_4$; anchor positions; sampling interval Δt; number of particles N

Initialization: Particles $\mathbf{p}_i = [x_i, y_i]^\top$ initialized around the LSE position estimate according to a chosen spatial spread;

Weights initialized uniformly: $w_i = 1/N$;
Process noise $\mathbf{Q}_{\text{pf}}$ (tunable, consistent with position units);
Measurement noise parameter(s) defining the likelihood model (tunable);

while *measurement* $\mathbf{z}$ *available* **do**
  if *not initialized* **then**
    Compute initial position using LSE;
    Distribute particles around this position;
    Mark initialized;
  end
  `// prediction`
  Propagate each particle using a motion model and add noise $\sim \mathcal{N}(0, \mathbf{Q}_{\text{pf}})$;
  `// update`
  Compute importance weights from measurement likelihood;
  Normalize weights;
  `// resampling`
  Compute $N_{\text{eff}} = 1/\sum_i w_i^2$;
  if N_{eff} *below threshold* **then**
    Resample particles; reset $w_i = 1/N$;
  end
end

Output : Estimated position $[x, y]^\top$

4.2 System evaluation

The system uses four fixed UWB anchors and one mobile tag. The anchor coordinates (mm) are: $\mathbf{A}_1 = (0, 0)$, $\mathbf{A}_2 = (5470, 0)$, $\mathbf{A}_3 = (5420, 5050)$, and $\mathbf{A}_4 = (770, 5050)$. With this layout, the test area is approximately 5.5 m $\times$ 5 m under LOS conditions.

The robot moves sequentially at 0.5 m/s through points $\mathbf{A} = (1200, 2710)$, $\mathbf{B} = (1200, 600)$, $\mathbf{C} = (4800, 600)$, $\mathbf{D} = (4800, 2710)$, and returns to $\mathbf{A}$, forming a closed rectangular trajectory.

Overview of scenarios: Two complementary scenarios are executed.

Scenario 1 — PF computational cost vs. particle count.

This scenario evaluates how the number of particles N affects PF runtime and positioning performance. The robot repeats the fixed trajectory while N is varied across trials. For each configuration we record:

- per-step processing time (averaged and median),
- achieved sampling rate (Hz),
- estimation error samples and empirical CDF,

Scenario 2 — Sensitivity of EKF/UKF to Q and R.

We analyze how process and measurement noise affect EKF and UKF performance. Both filters share the

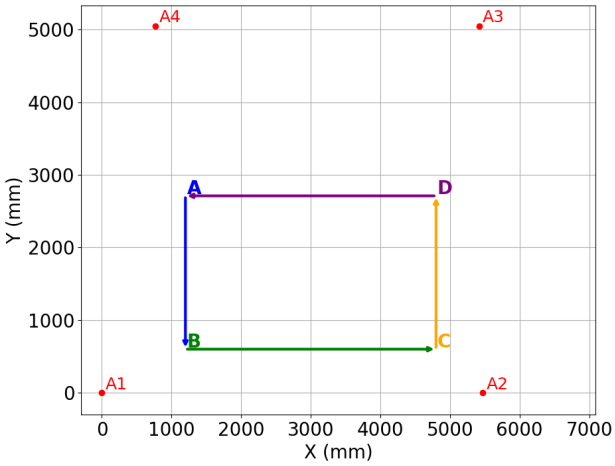

Figure 2. Experimental trajectory.

same initialization and baseline noise settings before sweeping:

$$\text{Baseline:} \quad \sigma_{a,0}^2 = 1, \quad \sigma_{r,0}^2 = 1.$$

A two-stage, one-at-a-time sweep is performed:

1) Fix $\mathbf{R} = \sigma_{r,0}^2 \mathbf{I}$ and sweep $\sigma_a^2 = s_Q \cdot \sigma_{a,0}^2$ over $s_Q \in \{10^{-5}, 10^{-4}, \ldots, 10^4\}$ (logarithmic grid).
2) Fix $\mathbf{Q} = \sigma_{a,0}^2 \mathbf{Q}_{\text{cv}}(\Delta t)$ and sweep $\sigma_r^2 = s_R \cdot \sigma_{r,0}^2$ over $s_R \in \{10^{-5}, 10^{-4}, \ldots, 10^4\}$.

The LSE (bias-calibrated) and the PF benchmark from Scenario 1 are included in all comparisons.

Parameter initialization and pragmatic prescriptions: For reproducibility and clarity we follow these prescriptions:

- **Initial state:** initial position from LSE trilateration; initial velocity set to zero.
- **Initial covariance $\mathbf{P}_0$:** diagonal matrix with variances that reflect the expected LSE spread and prior velocity uncertainty.
- **Process noise Q (EKF/UKF):** use the structured CV matrix $\mathbf{Q}_{\text{cv}}(\Delta t)$ and sweep only the scalar multiplier σ_a^2 as described above.
- **Measurement noise R:** modeled as diagonal; when anchors share similar characteristics use $\mathbf{R} = \sigma_r^2 \mathbf{I}$ and sweep via the scalar multiplier σ_r^2.
- **PF settings:** particle spread σ_{init}, resampling threshold (e.g., N_{eff} threshold), and $\mathbf{Q}_{\text{pf}}$ are selected and documented; these are tuned in Scenario 1.
- **Reproducibility:** fix and record random seeds; save exact $\mathbf{Q}$ and $\mathbf{R}$ matrices used for each run.

Metrics and reporting: For each trial and configuration we compute and report:

- **RMSE** over the whole trajectory,
- **Empirical CDF** of per-sample Euclidean positioning error with key percentiles (50%, 90%, 95%),
- **Convergence traces:** error vs. time to assess transient behaviour,
- **Computational metrics** (PF: per-step runtime and achieved sampling rate; EKF/UKF: average update time),
- **Statistical summaries:** mean and standard deviation across M trials; shaded error bands or error bars in plots.

Selection of PF benchmark $(N^\star, \mathbf{Q}_{\text{pf}}^\star)$*:* From Scenario 1 choose the smallest particle count $N^\star$ that achieves RMSE within a tolerance within 5% of the best observed RMSE while maintaining per-step runtime compatible with real-time operation. Use this tuned PF as the benchmark in Scenario 2.

The above setup ensures unit consistency, isolates the effect of process vs. measurement noise, and produces statistically robust comparisons between EKF, UKF, LSE and PF.

4.3 Results

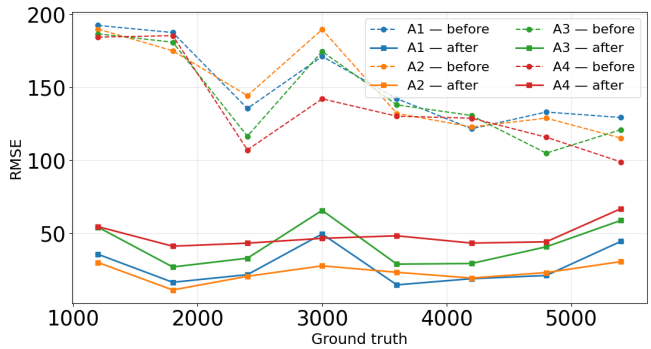

Figure 3. Bias calibration

4.3.1 Calibration results: Figure 3 shows the RMSE values of four anchors at different ground-truth distances under LOS conditions, comparing results before and after calibration. Before calibration, the RMSE is relatively high (around $100 - 190$ mm) and varies significantly with distance, indicating systematic bias and measurement noise in the raw UWB data. After calibration, the RMSE decreases considerably to about $15 - 65$ mm and becomes more stable. This result demonstrates that the calibration process effectively reduces systematic errors, such as hardware delays and antenna-related timing offsets in the SDS–TWR process, thereby improving the overall accuracy of the system.

Fig. 4 presents the CDF of the positioning errors obtained using the conventional LSE method and the calibrated LSE approach.

It can be observed that the CDF curve of the calibrated LSE consistently lies above and to the left of the raw LSE curve. This indicates that, for any given error threshold, a larger proportion of samples exhibit smaller positioning errors after calibration.

At an error threshold of approximately 100 mm, the calibrated method achieves a cumulative probability of about $0.8 - 0.85$, whereas the raw LSE only reaches around $0.65 - 0.7$. This demonstrates that nearly 80% of the calibrated estimates fall below 100 mm, compared to roughly 65% for the uncalibrated case.

The improvement is further confirmed by the RMSE values, which decrease from 111.6 mm (raw) to 72.7 mm

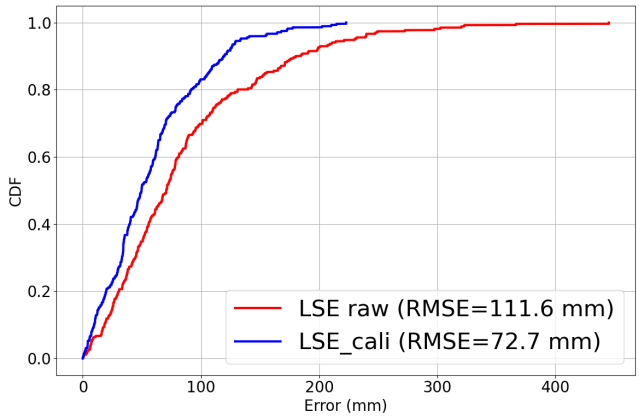

Figure 4. CDF LSE raw and calibrated

(calibrated), corresponding to an error reduction of approximately 35%. Additionally, the maximum observed error is significantly reduced after calibration, indicating that the proposed calibration model effectively mitigates systematic bias and suppresses large-error outliers.

Overall, the results clearly demonstrate that bias calibration substantially enhances the accuracy and robustness of the LSE-based positioning system.

4.3.2 Evaluation results:

Scenario 1:

Table I
Sampling Rate

Number of particles	10	30	50	70	90
Number of samples	4112	3688	3247	2800	2595
Sampling rate (Hz)	128	119	108	94	87

Table II
RMSE comparison

	Particle number	RMSE (mm)	Improvement vs. LSE
Raw	10	**83.1**	**25.53**%
	30	106.6	4.48%
	50	100.8	9.67%
	70	107.5	3.67%
	90	109.8	1,61%
Calibrated	10	**56.9**	**21.73**%
	30	64.5	11.28%
	50	73.8	−1.51%
	70	76.3	−4.95%
	90	63.9	12.1%

From Table I and Table II, it can be observed that the number of particles significantly affects both computational efficiency and localization accuracy.

As shown in Table I, when the particle number increases from 10 to 90, the sampling rate gradually decreases from 128 Hz to 87 Hz. Moreover, the number of generated samples is reduced considerably (from 4112 to 2595). This indicates that increasing the number of particles increases computational complexity, thereby reducing the update frequency of the system.

Regarding localization accuracy, Table II presents the RMSE results for both raw and calibrated data. In the raw scenario, the best performance is achieved with 10 particles, resulting in an RMSE of 83.1 mm. When the particle number increases to 30, 50, 70, and 90, the RMSE increases to 106.6 mm, 100.8 mm, 107.5 mm, and 109.8 mm, respectively. This demonstrates that increasing the particle number does not improve accuracy in the raw case; instead, it may even degrade the estimation performance.

In the calibrated scenario, the overall RMSE values are significantly reduced. The lowest RMSE is again obtained with 10 particles, reaching 56.9 mm. When the particle number increases, the RMSE rises to 64.5 mm (30 particles), 73.8 mm (50 particles), and 76.3 mm (70 particles), before slightly decreasing to 63.9 mm at 90 particles. However, none of these configurations outperform the 10-particle case.

This behavior can be explained by the trade-off between the number of particles and the update speed of the filter. In theory, using more particles can provide a better approximation of the posterior distribution. However, in this experiment, increasing the particle number greatly reduces the sampling rate, which means the filter is updated less frequently within a fixed time period. As a result, when more than 10 particles are used, the particle filter does not have enough updates to converge properly in each motion segment, leading to larger estimation errors and higher RMSE values.

On the other hand, when using 10 particles, the particle filter operates at a much higher update rate. This allows faster weight updates and more frequent resampling. Together with the low-noise environment and the bias-calibrated measurements, the filter can make better use of temporal information and converge more reliably, even with a smaller number of particles.

Overall, the results indicate that increasing the particle number does not necessarily lead to better localization accuracy. On the contrary, a smaller number of particles (10 particles in this scenario) achieves the best trade-off between accuracy and computational efficiency, providing both the highest sampling rate and the lowest RMSE in both raw and calibrated cases.

Scenario 2:

Fig. 5 illustrates the RMSE performance of the EKF and UKF under different values of the process noise covariance Q prior to bias calibration. For reference, the dashed horizontal lines represent the RMSE of PF with 10 particles (83.1 mm) and the LSE baseline (111.6 mm).

It can be observed that the EKF is highly sensitive to the selection of Q. When Q is very small (10^{-5}), the RMSE reaches 329.0 mm, indicating severe model mismatch due to underestimation of process uncertainty. As Q increases, the EKF performance improves significantly, reaching a minimum RMSE of 51.3 mm at

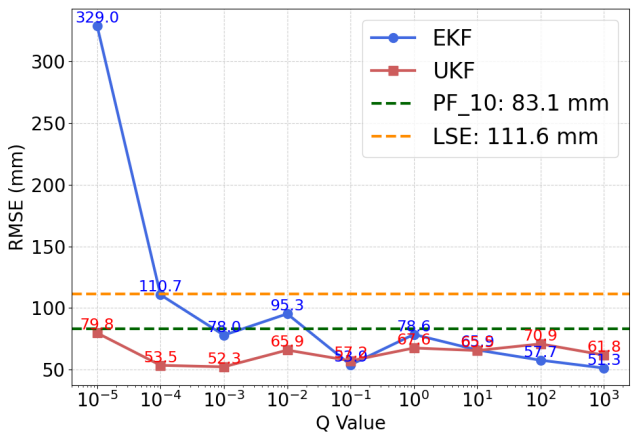

Figure 5. RMSE when varying the process noise covariance Q (raw measured)

$Q = 10^3$. However, noticeable fluctuations occur across intermediate Q values, demonstrating that improper tuning may degrade stability.

In contrast, the UKF exhibits more stable behavior across the entire Q range. The RMSE remains within a relatively narrow interval between approximately 52 mm and 80 mm. The best performance is achieved at $Q = 10^{-3}$ with an RMSE of 52.3 mm. Compared to EKF, the UKF shows reduced sensitivity to Q variations, indicating stronger robustness against process noise mis-specification.

Furthermore, for most Q values, both EKF and UKF outperform the LSE baseline (111.6 mm) and the PF method with 10 particles (83.1 mm). This confirms the advantage of recursive Bayesian filtering over batch least-squares estimation in handling measurement noise and dynamic uncertainty.

Overall, the results highlight the critical importance of proper Q tuning, especially for the EKF. While EKF can achieve competitive accuracy with optimal tuning, UKF provides more consistent performance across a wide range of process noise settings.

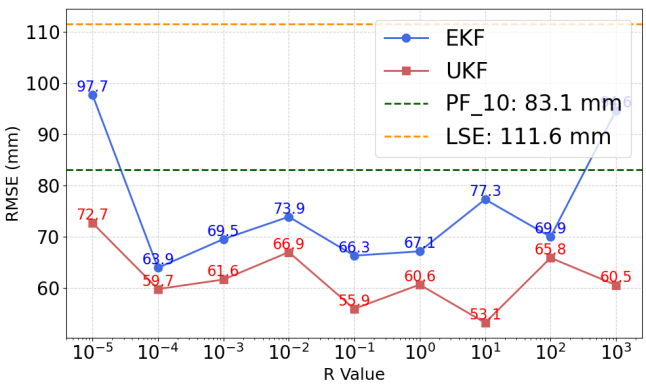

Figure 6. RMSE when varying the measurement noise covariance R (raw measured)

Fig. 6 presents the RMSE performance of the EKF and UKF under different values of the measurement noise covariance R prior to bias calibration. The dashed horizontal lines denote the RMSE of PF with 10 particles (83.1 mm) and the LSE baseline (111.6 mm).

It can be observed that both EKF and UKF exhibit relatively stable behavior across a wide range of R values compared to the sensitivity observed with Q tuning. For EKF, the RMSE varies between approximately 63.9 mm and 97.7 mm. The best performance is achieved at $R = 10^{-4}$ with an RMSE of 63.9 mm, while overly small measurement noise ($R = 10^{-5}$) results in degraded performance (97.7 mm), likely due to overconfidence in noisy measurements.

Similarly, the UKF demonstrates consistent performance, with RMSE values ranging from approximately 53.1 mm to 72.7 mm. The optimal performance occurs at $R = 10$, where the RMSE reaches 53.1 mm. Compared to EKF, UKF consistently achieves lower RMSE across most R settings, indicating stronger robustness to measurement noise covariance selection.

Furthermore, for nearly all tested R values, both EKF and UKF outperform the PF benchmark (83.1 mm) and significantly surpass the LSE baseline (111.6 mm). This confirms the effectiveness of recursive filtering methods in mitigating measurement noise.

Overall, the results indicate that proper tuning of R improves estimation accuracy, but both filters remain relatively robust over a broad range of measurement noise settings. Among them, UKF demonstrates superior stability and consistently better accuracy under the non-calibrated condition.

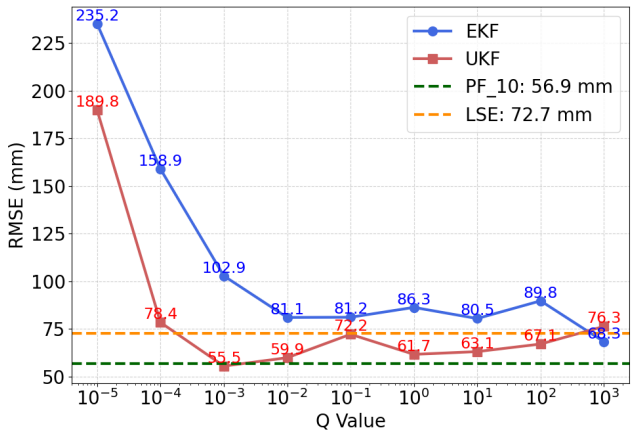

Figure 7. RMSE when varying the process noise covariance Q (calibrated measured)

Fig. 7 shows the RMSE results of the EKF and UKF for different values of the process noise covariance Q after bias calibration. The dashed horizontal lines represent the calibrated LSE result (72.7 mm) and the PF with 10 particles (56.9 mm).

After calibration, both filters become more stable than in the non-calibrated case. For the EKF, the RMSE is very high when Q is small. At $Q = 10^{-5}$, the RMSE reaches 235.2 mm, and it decreases to 158.9 mm at $Q = 10^{-4}$. When Q increases to moderate values, the EKF performance improves clearly. The RMSE is 102.9 mm at $Q = 10^{-3}$ and drops further to 81.1 mm at $Q = 10^{-2}$. The best EKF result is achieved at $Q = 10$ with an RMSE of 80.5 mm. However, when Q becomes

large, the RMSE increases again to 89.8 mm at $Q = 10^2$ before decreasing to 68.3 mm at $Q = 10^3$. This shows that the EKF performance strongly depends on the choice of Q.

The UKF shows more robust behavior across the tested Q range. At very small Q, the RMSE is 189.8 mm for $Q = 10^{-5}$ and decreases to 78.4 mm at $Q = 10^{-4}$. The best UKF performance is obtained at $Q = 10^{-3}$, where the RMSE reaches 55.5 mm. For moderate values of Q, the RMSE remains relatively stable, with values of 59.9 mm at $Q = 10^{-2}$, 72.2 mm at $Q = 10^{-1}$, 61.7 mm at $Q = 1$, and 63.1 mm at $Q = 10$. Even at larger Q values, the RMSE stays below 76.3 mm.

At its optimal point ($Q = 10^{-3}$), the UKF achieves an RMSE of 55.5 mm, which is slightly better than the PF result with 10 particles (56.9 mm) and clearly better than the LSE baseline (72.7 mm). Overall, the UKF is less sensitive to changes in Q than the EKF.

Overall, the results confirm that calibration not only improves absolute positioning accuracy but also enhances filter stability. While EKF requires careful tuning of Q to avoid divergence, UKF provides more consistent and reliable performance across a broad range of process noise settings.

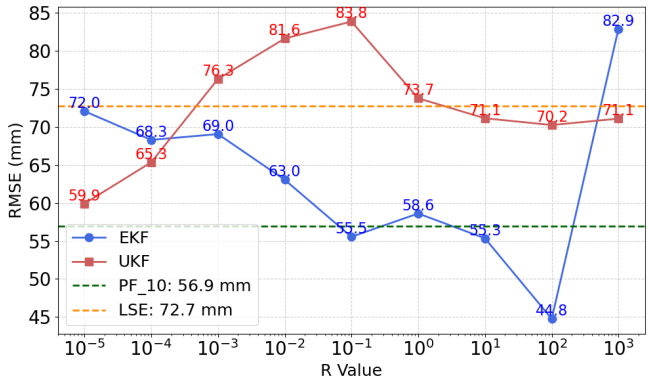

Figure 8. RMSE when varying the measurement noise covariance R (calibrated measured)

Fig. 8 presents the RMSE performance of the EKF and UKF under different values of the measurement noise covariance R after bias calibration. The dashed horizontal lines indicate the calibrated LSE baseline (72.7 mm) and the PF method with 10 particles (56.9 mm).

It can be observed that both filters demonstrate improved stability compared to the non-calibrated case. For EKF, the RMSE ranges from 44.8 mm to 82.9 mm. The optimal performance is achieved at $R = 10^2$, where the RMSE reaches a minimum of 44.8 mm, significantly outperforming both the calibrated LSE (72.7 mm) and PF (56.9 mm) benchmarks. However, excessively large measurement noise ($R = 10^3$) leads to performance degradation (82.9 mm), indicating overestimation of measurement uncertainty.

In contrast, the UKF exhibits relatively moderate variation across the tested R values. The RMSE ranges between approximately 59.9 mm and 83.8 mm. The best UKF performance is observed at $R = 10^{-5}$ with an RMSE of 59.9 mm. Although UKF does not achieve as low a minimum RMSE as EKF in this case, it maintains more consistent behavior without sharp fluctuations.

Furthermore, for most moderate R values (10^{-4} to 10^2), the EKF consistently outperforms UKF after calibration. This suggests that once systematic bias is compensated, the EKF can achieve superior accuracy provided that the measurement noise covariance is properly tuned.

Overall, the results indicate that after calibration, tuning the measurement noise covariance R plays a crucial role in achieving optimal estimation accuracy. While UKF maintains stable performance across a broad range of R settings, the EKF demonstrates the potential for significantly lower RMSE when an appropriate R value is selected.

5 Conclusion and future work

In this study, we have evaluated the applicability of EKF, UKF, and PF for UWB-based indoor robot positioning with four fixed anchors and one mobile tag. The proposed bias calibration effectively removed the dominant systematic ranging offset (approximately −140 mm) and improved the LSE positioning RMSE from 111.6 mm (raw) to 72.7 mm (calibrated), while also reducing large-error outliers. In Scenario 1, PF performance was shown to depend strongly on the number of particles, revealing a clear trade-off between computational cost (sampling rate) and estimation quality. In Scenario 2, varying Q and R significantly affected EKF/UKF convergence; after calibration, both filters consistently outperformed the LSE baseline, with EKF achieving very low RMSE when R was appropriately tuned and UKF providing more robust performance across a wider range of covariance settings.

For future work, the accuracy and robustness of the proposed system can be further enhanced by integrating learning-based state estimation and simultaneous localization and mapping techniques.

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